Huawei Access Node Routing Protocol Flap Failure Resolution SOP (KR-ACC-72)

Node: KR-ACC-72 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_6631

Description

The KR Puram Access Node, a Huawei Access node (Node ID: KR-ACC-72), has experienced a Routing Protocol Flap failure, resulting in a Critical alarm (Alarm Code: ALARM_6631). This failure domain is attributed to the dynamic nature of BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. In this scenario, the Access node, running the Huawei CLI dialect, has encountered a situation where the routing protocols are experiencing a high rate of convergence, leading to a temporary loss of routing table consistency. This, in turn, causes the node to enter a state of instability, triggering the Critical alarm. The underlying protocol mechanics, including BGP's path-vector routing, OSPF's link-state routing, and ISIS's link-state routing, contribute to this failure domain. Furthermore, the Access node's hardware, including the routing engine and line cards, plays a crucial role in processing the high volume of routing protocol updates, which can lead to packet loss and increased latency. The combination of these factors necessitates a thorough investigation and corrective action to prevent future occurrences.

Pre-requisite Checks

- Verify that the KR Puram Access Node (KR-ACC-72) is accessible and can be reached remotely.

- Confirm that the redundant link is operational and can assume traffic in the event of a primary link failure.

- Check the environmental conditions at the KR Puram Access site, ensuring that temperature, humidity, and power levels are within the recommended ranges.

Potential Root Causes

Cause	Probability
BGP Route Churn	High
OSPF Adjacency Loss	Medium
ISIS LSP Flooding	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp routing-table, Expected Output: BGP routing table information, including the number of routes and the source of the update., Description: This command displays the BGP routing table, providing insight into the current state of the routing protocol, including the number of routes and the source of the update.
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```
Command: display ospf neighbor, Expected Output: OSPF neighbor information, including the neighbor ID and state., Description: This command displays the OSPF neighbor information, including the neighbor ID and state, to help diagnose adjacency loss issues.

Command: display isis lsp, Expected Output: ISIS LSP information, including the LSP ID and metric., Description: This command displays the ISIS LSP information, including the LSP ID and metric, to help diagnose LSP flooding issues.
```

Resolution Steps

Step 1: Step: Reset BGP Session, Command: reset bgp restart, Description: Reset the BGP session for the specified peer group to resolve BGP route churn issues.

Step 2: Step: Re-establish OSPF Adjacency, Command: clear ospf process, Description: Clear the OSPF process and re-establish the adjacency to resolve OSPF adjacency loss issues.

Step 3: Step: Filter ISIS LSPs, Command: isis filter-lsp, Description: Filter ISIS LSPs to prevent excessive LSP flooding and resolve ISIS LSP flooding issues.

Post-Check Verification

```
Command: display bgp routing-table, Expected Output: Stable BGP routing table with no churn., Description: Verify that the BGP routing table is stable and there is no churn.

Command: display ospf neighbor, Expected Output: Stable OSPF neighbor state., Description: Verify that the OSPF neighbor state is stable.

Command: display isis lsp, Expected Output: Stable ISIS LSP information., Description: Verify that the ISIS LSP information is stable.
```

Roll-back Procedures

- Step: Revert BGP Changes, Command: undo reset bgp , Description: Revert the BGP changes made during the resolution process.

- Step: Revert OSPF Changes, Command: undo clear ospf process, Description: Revert the OSPF changes made during the resolution process.

- Step: Revert ISIS Changes, Command: undo isis filter-lsp, Description: Revert the ISIS changes made during the resolution process.